NWAV54 Project Launch

# Perceptual accessibility of the Ontario-Quebec French dialect boundary

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## BACKGROUND

On paper:

Laurentian French has historically been treated as a single, phonetically **homogenous** variety.
(Léon, 1968)

In experience:

Franco-Ontarians report being **misidentified** by Québécois;
L2 learners describe Ontario French as more **intelligible**.

What might drive regional differences?

**Language restriction**
(Mougeon & Béniak, 1991; Mougeon, 1997; Thomas, 1986)

**Bilingualism**
(Papen, 1998)

**Milieu**
(Remysen, 2016)

Direct phonetic comparisons of Ontario French and Quebec French show mixed results.

**Shared variation**

- /l/-elision (Poplack & Walker, 1986)
- Nasalization (Séguinot, 1968, cited in Thomas, 1989)
- Loanword adaptation (Poplack et al., 2020)

**Divergences**

- VOT (MacLeod, 2006)
- /ə/-deletion (Poiré et al., 2010; Uritescu et al., 2004)
- Articulation rate (Cichocki et al., 2024)

## RESEARCH QUESTIONS

1. Can listeners perceive **fine-grained phonetic regional variation** in Laurentian French? If so, are perceptual boundaries structured by **province**, **contact** with English, or **milieu**?
2. How does listener background shape regional perception?
3. Is regional identification driven by acoustic variation?

## PRELIMINARY (PILOT) RESULTS

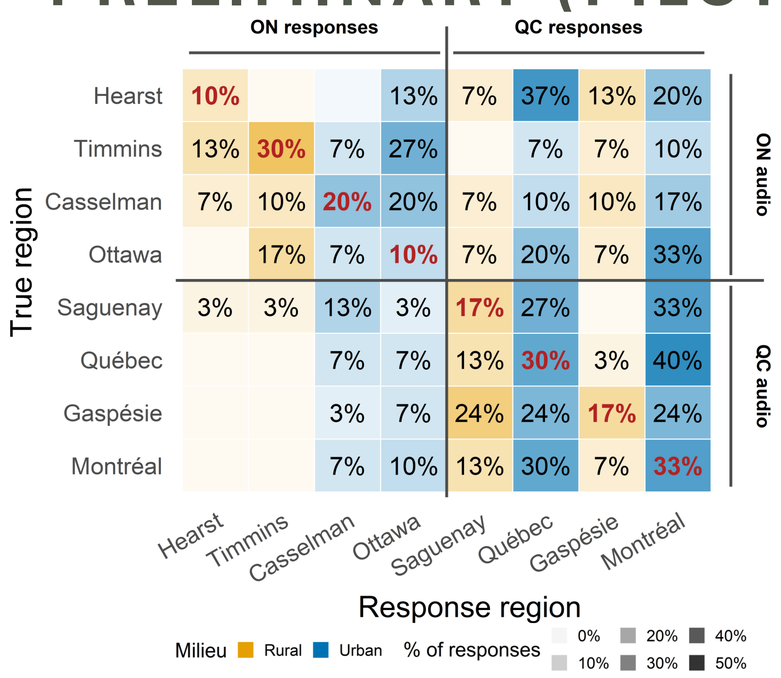



Figure 1: Confusion matrix for True region × Response region.
Chance = 12.5% (1/8 regions); n = 10.

**Region-level identification is weak above chance: Timmins, Montréal, Québec**

**Errors cross ON → QC but rarely the other way**

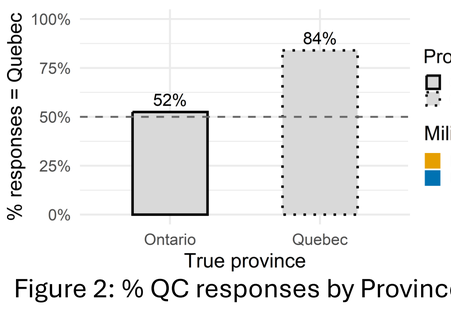



Figure 2: % QC responses by Province

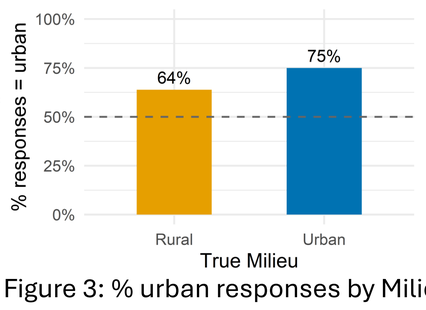



Figure 3: % urban responses by Milieu

**Bias towards QC & urban responses**

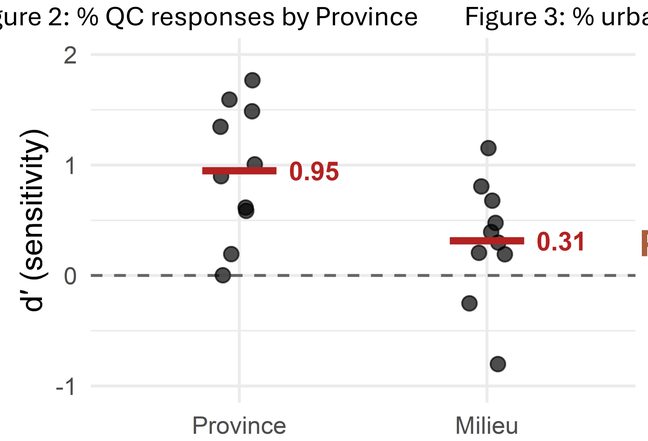



Figure 4: Sensitivity (d') to Province & Milieu.
Each point is one listener, n = 10; red line = mean, dashed line = chance.

**Province is easier to hear than Milieu**

## METHOD

**PARTICIPANTS**

**10 Listeners**

- 8 L1–French, 2 L2–French (frequent use)
- 7 Montreal, 1 Gatineau, 1 Ottawa (1 NA)
- 6 women, 3 men (1 NA)
- 22–68 years old (median: 38)

**STIMULI**

**48 Stimuli = 8 Regions × 6 Speakers**

- L1–French
- Born/raised in each region
- Spontaneous speech (publicly available)
- ~20s audio clips

| Province | Region | L1 French pop. | Speakers |
|---|---|---|---|
| Ontario | Hearst | 88.4% | F32, F43, F73, M34, M45, M68 |
| | Casselman | 75.6% | F33, F38, F39, M29, M64, M68 |
| | Timmins | 37.2% | F30, F41, F79, M30, M55, M60 |
| | Ottawa | 17.6% | F21, F26, F54, F73, M19, M72 |
| Quebec | Saguenay | 97.5% | F28, F46, M43, M44, M56, M69 |
| | Québec (City) | 92.4% | F31, F50, F58, M19, M51, M82 |
| | Gaspésie | ~90% | F33, F41, F62, M45, M66, M78 |
| | Montréal | 59.9% | F24, F38, F76, M29, M55, M78 |

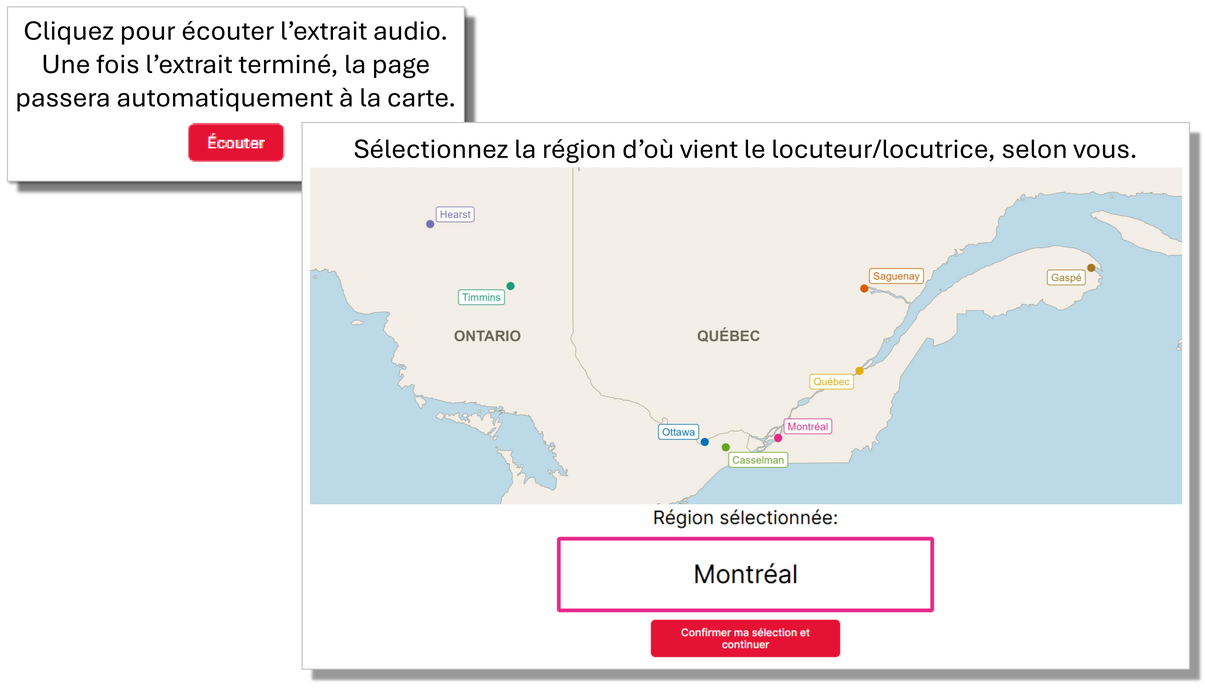
Table 1: Percentages of L1 French speakers by region (Statistics Canada, 2021; Statistique Québec, 2021) in descending order, colour-coded by milieu (urban vs. rural).

**TASK**

Map-based **8AFC**

(Clopper & Pisoni, 2004; Preston, 1993; Remysen, 2016)

**2 Practices** + **24 Trials** (random 3/6 per region)




### Closing questions

**Quels indices ont influence vos choix?**

- ❑ Intuition
- ❑ Voyelles
- ❑ Consonnes
- ❑ Vitesse
- ❑ Mot/expressions
- ❑ Ton/intonation
- ❑ Autres: _____

**Avez-vous utilisé une stratégie particulière?**

- ☑ Oui
- ☒ Non

Si oui, laquelle? _____

**Historique:**

- Informations démographiques
- Antécédents linguistiques

## NEXT STEPS

### REFINE THE TASK

**8AFC too hard**

- Fewer options
- Free classification task (Clopper & Pisoni, 2007)

**Scale up**

- Larger stimuli pool (more speakers, more clips)
- More listeners (n = 40, ON & QC)

### TOWARDS FULL RESULTS

**Analysis plan**

- Signal detection theory (sensitivity vs. bias)
- Confusion matrices + Similarity Choice Model (Clopper & Pisoni, 2004)
- Bayesian regression with listener demographics and language background, speaker effects…
- Acoustic analysis of stimuli

Revise task → Full data collection → Analysis & modelling → Acoustics

**Feedback welcome!**

## ACKNOWLEDGEMENTS & REFERENCES

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